

IronLot: Automated Bidding Model for Used Car Auctions

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Task 1 Report · Pawn Stars

Data Cleaning & Wrangling

Dataset Overview and Train-Validation Split

The raw dataset contained 447,048 rows across 12 columns. All column dtypes were appropriate. The very first action before any cleaning was a stratified 80/20 train-validation split (`random.state=42`): every imputation statistic, lookup table, and encoder was computed on the training partition only, then applied identically to validation. No information from the validation set influenced any cleaning decision.

Missing Value Analysis

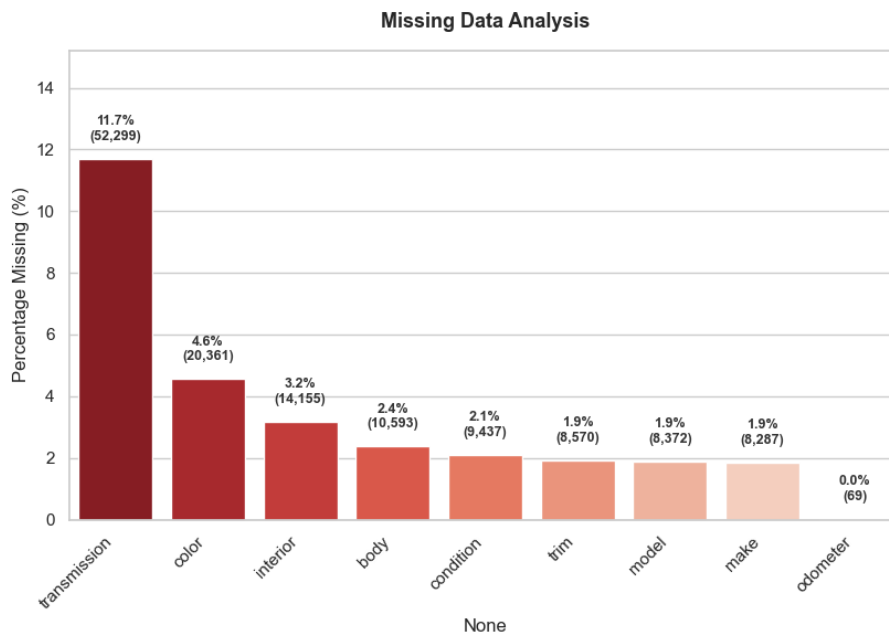


Figure 1: Missing value analysis across all 12 columns. `transmission` has the highest rate at 11.7%; `year` and `sellingprice` have zero missing values.

Missing rates ranged from 0.02% (`odometer`) to 11.7% (`transmission`). A structural dependency was discovered: whenever `make` was missing, `model` and `trim` were also missing in 100% of cases — confirmed by checking intersection cardinalities. Since `make`, `model`, and `trim` are the dominant drivers of selling price, imputing them would introduce severe false signal. An agent that cannot identify a car's make or model cannot meaningfully bid on it. These 8,287 rows (< 2%) were dropped — the only rows removed from the entire dataset.

Hierarchical Imputation Strategy

For all remaining missing columns a three-tier hierarchical strategy was used, falling back from most specific to most general context:

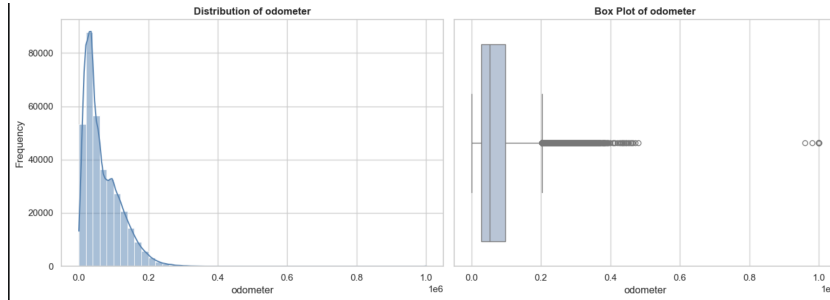
Transmission (11.7% missing): Grouped missingness analysis by body type showed elevated rates for e-series vans and CTS coupe, classifying this as Missing At Random (MAR). Imputation order: (1) mode for exact `make+model` pair, (2) mode for `make`, (3)

global mode (`automatic`). Lookup tables computed on training data only and saved to encoders.

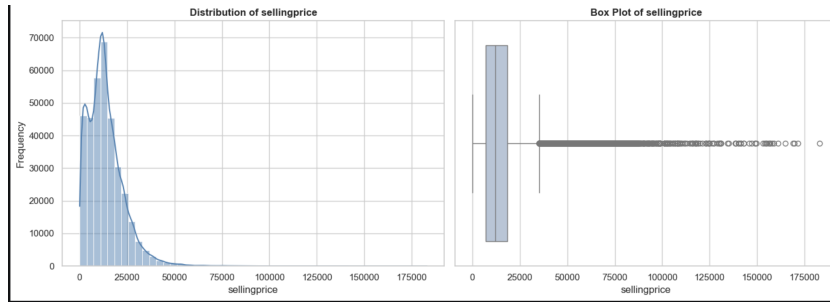
Odometer and Condition (0.02% and 2.1%): Hierarchical median: (1) median for `year+model`, (2) median for `year`, (3) global median. A critical bug was caught: for rare (`year`, `model`) pairs where every training row was also missing the target column, Pandas computes a median of `NaN`. A `pd.notna()` guard was added at every tier to bypass invalid entries and cascade correctly.

Body, Color, Interior: Mode hierarchy at (1) `make+model+trim`, (2) `make+model`, (3) global mode — these attributes are largely determined by a car’s exact trim specification.

Outlier Detection and Treatment



(a) Odometer distribution with log-IQR upper bound (\$633,821 mi) marked in red.



(b) Selling price distribution with log-IQR upper bound (\$77,347) marked in red.

Figure 2: Log-IQR outlier bounds for the two skewed numeric features. Standard IQR produced negative lower bounds; log transformation corrects for right skew before applying the rule.

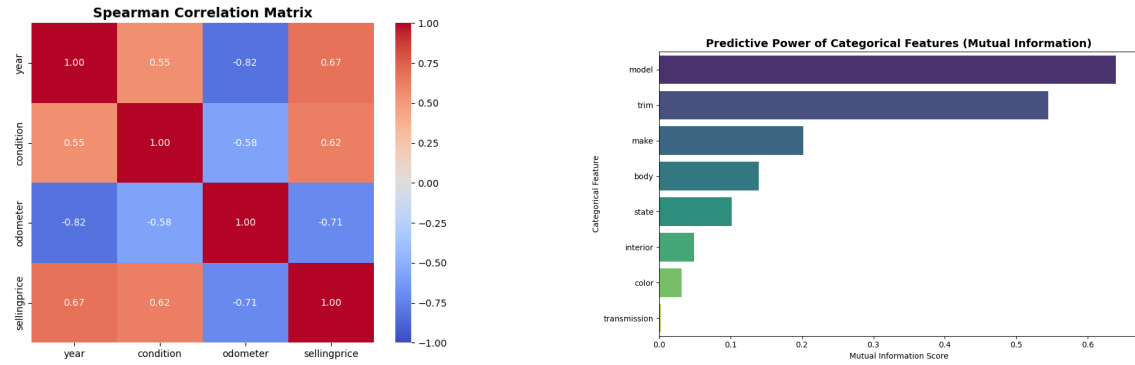
Standard IQR on raw `sellingprice` produced a nonsensical upper bound of \$35,250 and a negative lower bound — a direct consequence of applying a symmetric method to a right-skewed distribution (skewness > 2.1 , excess kurtosis > 15). The fix: apply IQR on the log-transformed distribution, then exponentiate:

$$\text{upper} = \exp(Q_3(\ln p) + 1.5 \cdot \text{IQR}(\ln p)) - 1 \quad (1)$$

This yielded bounds of \$1,655–\$77,347 for price and \$4,345–\$633,821 for odometer. High-price outliers ($> \$77k$) were confirmed genuine: top makes were Mercedes-Benz, Porsche, Ferrari, Rolls-Royce, mean odometer 12,150 miles, mean condition 4.45 — retained entirely. Low-price outliers ($< \$1,655$): mean odometer 158,094 miles, mean condition 2.07 — genuine salvage vehicles, also retained. For odometer, values < 10 or $> 900,000$ (data entry errors) were set to `NaN` and re-imputed; values above the log-IQR bound were kept intact.

Exploratory Data Analysis

Bivariate Correlation Analysis



(a) Spearman rank correlation matrix. Used over Pearson due to right-skewed, non-normal distributions in `odometer` and `sellingprice`.

(b) Mutual Information scores for categorical features. `model` and `trim` dominate; `transmission` carries near-zero marginal signal.

Figure 3: Numeric and categorical feature importance analysis.

Key Spearman findings: `odometer`→`price` (-0.71 , strongest numeric predictor), `year`→`price` ($+0.67$), `condition`→`price` ($+0.62$), and `year`↔`odometer` (-0.82 , multicollinearity). PCA was considered to resolve the year–odometer collinearity but rejected: it would collapse both into an uninterpretable PC1. Both features were retained since LightGBM handles collinearity through independent feature selection at each split and both carry independent pricing signal. MI scores confirmed `model` (0.639) and `trim` (0.545) are the most informative categoricals, more so than `make` (0.201) alone.

Depreciation Across Body Types

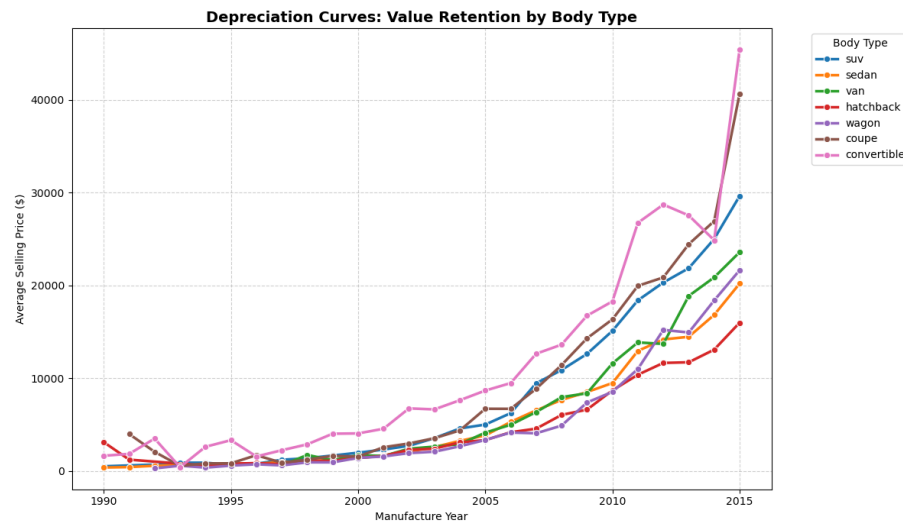
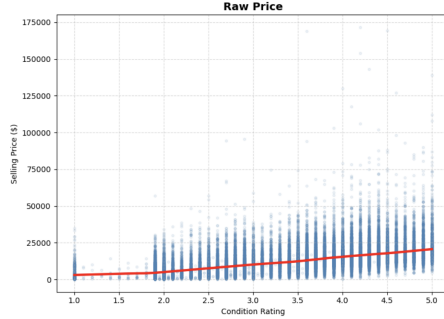


Figure 4: Depreciation curves by body type. Coupe and convertible diverge sharply post-2010 (composition effect). All body types converge below \$2,000 pre-1999. Van separates from SUV after 2008 (demand shift).

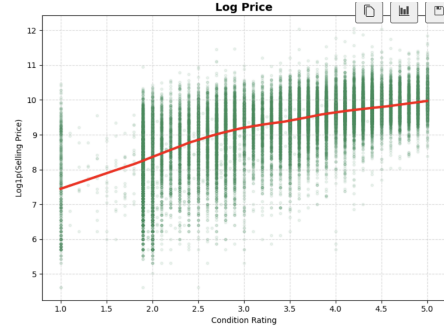
Three non-obvious patterns emerged: (1) **Composition effect:** coupes reach \$40,500 by 2015 not because coupes appreciated, but because the mix shifted from economy coupes (pre-2005) to luxury/sports vehicles (BMW 4-series, Porsche 911) post-2010. (2) **Pre-2000 convergence:** all body types collapse within \$500–2,000 before 1999 — body type loses virtually all predictive power for vehicles older than 15–20 years, with odometer and

condition dominating. (3) **Demand-side van depreciation:** vans tracked SUVs closely until 2008, then diverged by nearly \$10,000 as consumer preference shifted to crossovers, collapsing van residual values regardless of mechanical condition.

Condition Non-Linearity: Structural Break at 3.0

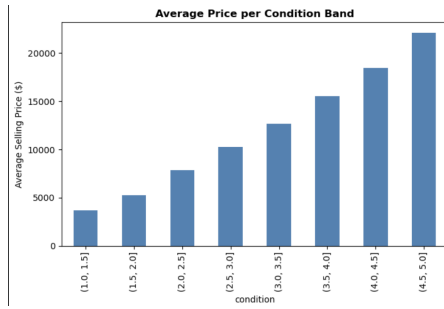


(a) Raw price: LOWESS curve dragged down by cheap car volume. Spread widens dramatically above condition 3.

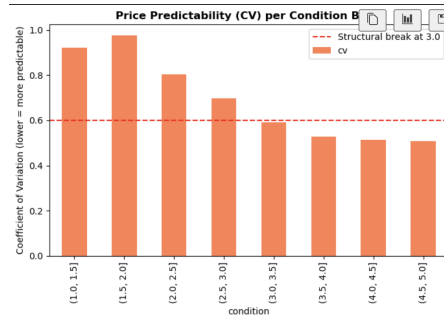


(b) Log price: reveals proportional gains are largest in condition 1–3 range. Curve flattens above 3.5.

Figure 5: LOWESS non-linearity analysis for condition vs selling price.



(a) Average price per condition band — monotonically increasing.

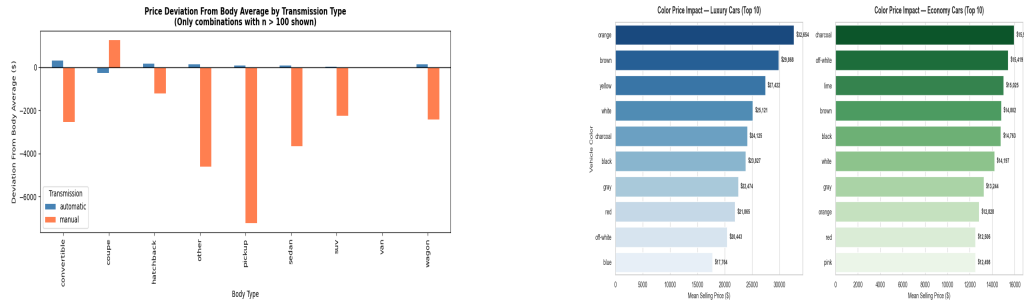


(b) CV per band. Structural break at condition 3.0: CV drops from 0.697 to 0.590.

Figure 6: Condition band analysis quantifying the wholesale-to-retail pricing threshold.

CV ($= \sigma/\mu$) per 0.5-point band reveals two structural breaks. First at condition 2.0: CV peaks at 0.975 because the band mixes \$200 scrap and \$22,000 hail-damaged BMWs — vehicle identity dominates condition score. Second at condition 3.0: CV drops from 0.697 to 0.590, marking the **wholesale-to-retail threshold** where cars become dealer-lot candidates, expanding buyer pools and stabilising prices. These thresholds directly set the agent's confidence boundaries ($\text{WIDTH_HIGH_CONF} = 0.139$, $\text{WIDTH_PASS} = 0.414$).

Transmission × Body Type and Color/Interior Analysis



(a) Transmission × body deviation. Manual coupes command \$1k premium; manual pickups trade \$7k below average. (b) Color price impact: \$15k spread in luxury vs \$3.5k in economy. Orange encodes exotic manufacturer, not color preference.

Figure 7: Conditional pricing effects for transmission-body combinations and color segments.

Transmission is 96.9% automatic. Manual coupes command $\approx$ \$1,000 above the coupe average (enthusiast premium); manual pickups trade $\approx$ \$7,000 below pickup average (commercial buyers require automatic). Color and interior, despite low marginal MI scores (0.031 and 0.049), show \$15k and \$13k spreads respectively within the luxury segment — both encode trim level and bespoke specification, not surface preferences. Binary interaction flags (`luxury_exotic_color`, `luxury_premium_interior`) were engineered to capture these conditional effects robustly.

Usage Intensity Versus Component Features

Usage intensity (odometer $\div$ car age) was hypothesised to outperform components. MI scores contradicted this: $MI(\text{year}) = 0.453$, $MI(\text{odometer}) = 0.420$, $MI(\text{usage_intensity}) = 0.066$. Division collapses two independent signals: a 2014 car with 10,000 miles and a 2004 car with 100,000 miles have identical usage intensity but dramatically different prices. Spearman correlation between `car_age` and `year` was exactly -1.0 — mathematically identical features. Both were dropped from model features; usage intensity is retained only in the agent for anomaly flagging.

Feature Engineering and Encoding

The raw `body` column’s 45 values were consolidated into 9 clean categories via a saved lookup dictionary (e.g., `crew cab`, `supercab`, `access cab` $\rightarrow$ `pickup`). **Target Encoding** (smoothing $\lambda = 10$) was applied to `make`, `model`, `trim`, and `state` via `category_encoders.TargetEncoder`, replacing each category with: $\hat{\mu}_c = (n_c \bar{y}_c + 10 \bar{y}_{\text{global}}) / (n_c + 10)$. Unseen categories return the global mean (\$13,699), verified empirically. **One-Hot Encoding** (`drop_first=True`) was applied to `transmission` and `body_clean`. Binary flags require no further encoding.

Model Selection, Training, and Hyperparameter Tuning

Model Selection: Why LightGBM Over Alternatives

Scatter plots of price against all numeric features showed non-linear, clustered relationships with conditional heteroscedasticity — immediately ruling out linear methods. **vs. Linear/Ridge/Lasso:** the interaction between `make`, `year`, and `condition` is fundamentally non-additive; a Ford from 1995 with 200,000 miles and a BMW from 2014 with 8,000 miles cannot be separated by any linear combination of raw features. **vs. Random Forest:** both are ensemble tree methods, but LightGBM builds trees *sequentially* — each new tree corrects prior residuals, directing capacity toward hard-to-predict examples. LightGBM also trains 5–10 $\times$ faster than Random Forest on 350k+ rows via histogram-based split finding rather than exact enumeration. **vs. XGBoost:** XGBoost grows trees *level-wise*

(breadth-first, all nodes at depth d before $d + 1$); LightGBM grows *leaf-wise* (best-first, the single highest-gain leaf at each step regardless of depth). Leaf-wise growth concentrates splits where gain is largest, achieving lower loss for the same number of leaves. The `num_leaves` parameter (optimally 267) gives direct complexity control that XGBoost’s `max_depth` alone cannot. Critically, LightGBM’s native `objective='quantile'` uses the same histogram infrastructure, enabling 25th and 75th percentile models at negligible additional training cost — this quantile architecture is the foundation of the agent’s uncertainty estimation and would require separate implementation in XGBoost. **vs. Neural Networks:** on tabular data under 500k rows, gradient boosted trees consistently match or outperform deep learning with substantially less engineering. Embedding layers for high-cardinality categoricals would partially replicate target encoding while introducing additional hyperparameters (embedding dimension, depth, batch size, learning rate schedule) each requiring their own tuning.

Regularisation and Overfitting Control

Three complementary mechanisms explain the tuned model’s 31% generalisation improvement. `subsample=0.661` trains each tree on 66.1% of data (stochastic gradient boosting), reducing variance. `colsample_bytree=0.560` randomly selects 56% of features per tree, decorrelating trees and forcing independent signal discovery. `min_split_gain=0.033` imposes a minimum loss-reduction threshold before any split is accepted, pruning nodes that memorise noise in rare make-trim combinations. Default LightGBM applies none of this regularisation.

Hyperparameter Tuning with Optuna TPE

Optuna with Tree-structured Parzen Estimator (TPE) was run for 100 trials with a Median Pruner (20 warm-up steps). An inner 90/10 split with 50-round early stopping was used inside each trial, independent of the outer validation set.

Parameter	Search Range	Type	Optimal
<code>num_leaves</code>	[20, 300]	int	267
<code>max_depth</code>	[4, 12]	int	12
<code>min_child_samples</code>	[20, 200]	int	20
<code>n_estimators</code>	[500, 3000]	int	2695
<code>learning_rate</code>	[0.005, 0.1]	log	0.04449
<code>subsample</code>	[0.6, 1.0]	float	0.6608
<code>colsample_bytree</code>	[0.5, 1.0]	float	0.5598
<code>reg_alpha</code>	[10^{-8} , 10]	log	7.83×10^{-8}
<code>reg_lambda</code>	[10^{-8} , 10]	log	0.6738
<code>min_split_gain</code>	[0.0, 1.0]	float	0.03295

The parameters tell a coherent story: 267 leaves with depth 12 permits deep, narrow trees capturing highly specific make-model-trim-condition combinations. Low `min_child_samples=20` allows small-subgroup leaves, critical for rare luxury vehicles (Ferrari, Bentley) that appear in small numbers but have dramatically different price behaviour. Near-zero `reg_alpha` with moderate `reg_lambda=0.674` indicates L1 sparsity was not needed — tree structure provides it — while L2 leaf-score regularisation stabilises predictions. Learning rate 0.0445 with 2,695 estimators follows the standard trade-off: slower learning with more trees generalises better.

Baseline vs. Tuned Comparison and Quantile Architecture

Model	Validation RMSE	vs. Baseline
Baseline (default LightGBM)	\$2,669.39	—
Tuned (Optuna, 100 trials)	\$1,841.91	−31.0% (\$827 improvement)

Both evaluated on the identical holdout. Tuned RMSE is 13.5% of mean price (\$13,663). Mean prediction bias = $-\$1.17$ with 49.7% underpredictions / 50.3% overpredictions — effectively zero systematic directional error. Two additional quantile models ($\alpha = 0.25$ and $\alpha = 0.75$) were trained with identical hyperparameters. Their spread forms the agent's uncertainty measure: $\text{relative_width} = (\hat{p}_{0.75} - \hat{p}_{0.25}) / \hat{p}_{0.50}$. Validation confirmed this independently recovers the CV finding: condition 1.0–1.5 cars give width 0.414; condition 4.5–5.0 give 0.087 — two independent methods, same conclusion.

Bidding Agent Design

Agent Pipeline

The agent loads three models at initialisation (relative paths for arena compatibility) and replicates the full preprocessing pipeline on each incoming single-row dictionary — no `df.mean()`, no `df.groupby()`, no training data reference. Every imputation rule, lookup table, and encoder is loaded at startup and accessed via dictionary lookup or fitted-object transform.

Core Philosophy and Bid Sequence Design

Since cars arrive sequentially, every pass is a permanently missed win. The strategy splits by confidence tier:

High confidence (width < 0.139 , from EDA structural break): maximise *profit per win* — reliable prediction, demand 10% margin, enter only if lower bound guarantees profit.

Low confidence (width $\in [0.139, 0.414)$): maximise *win count* — accept 5% margin, enter if mean shows any positive profit (wins weighted 20% vs. 15% for profit).

Step 1 — Confidence-tiered ceiling:

$$\text{anchor} = \begin{cases} \hat{p}_{0.50} & \text{high conf.} \\ \hat{p}_{0.25} & \text{low conf.} \end{cases} \quad \text{ceiling} = \text{anchor} \times (1 - m), \quad m \in \{0.10, 0.05\}$$

Anchoring to the 25th percentile for uncertain cars implements **bid shading**: in a first-price auction the winner is typically the most optimistic bidder (Winner's Curse); using the pessimistic quantile as ceiling systematically avoids overpaying.

Step 2 — Entry gates: Hard passes for model failure (predicted value $< \$100$; 38 validation cars with negative predictions had actual median price \$375 — abstaining costs nothing), extreme uncertainty (width ≥ 0.414), or ceiling below current bid. Conditional entry: high confidence requires lower-bound edge > 0 ; low confidence requires mean edge > 0 .

Step 3 — Fractional Kelly position sizing:

$$f_{\text{safe}} = \min\left(\frac{e}{\max(w^2, 10^{-6})} \cdot \frac{1}{2} \cdot \frac{B}{B_0}, f_{\text{max}}\right), \quad f_{\text{max}} = \begin{cases} 0.10 & \text{high conf.} \\ 0.04 & \text{low conf.} \end{cases}$$

where e is edge ratio, B/B_0 is bankroll ratio. Full Kelly is optimal over infinite bets but has dangerously high variance over finite sequences; half-Kelly reduces variance while sacrificing modest expected growth. The bankroll ratio B/B_0 provides automatic drawdown protection: if the bankroll falls to 60% of starting capital, every position automatically shrinks to 60% of its normal size — no explicit trigger required.

Step 4 — Hyperbolic decay bid sequence (deterministic):

$$b_{r+1} = b_r + \max\left((c_{\text{eff}} - b_r) \cdot f_{\text{safe}} \cdot \frac{1 - 0.3w}{1 + 0.5r}, \$50\right)$$

where r is the internal round counter (reset per car), $c_{\text{eff}} = \min(\text{ceiling}, b_r + B \cdot f_{\text{safe}})$, $w = \text{relative_width}$. Exponential decay γ^r collapses rapidly (at $\gamma = 0.8$, round 8 gives factor 0.17), causing tiny increments that signal budget exhaustion to rivals who then bid past the ceiling. Hyperbolic decay $1/(1 + 0.5r)$ decreases monotonically but slowly (round 8 $\rightarrow$ 0.20): bids remain credible throughout, preventing rivals from reading the agent’s proximity to its ceiling. The confidence penalty $(1 - 0.3w)$ additionally reduces increment speed for uncertain cars, exiting sooner when margin deteriorates. The bid is a pure function of $(\hat{p}_{0.50}, \hat{p}_{0.25}, B, b_{\text{curr}}, r)$ — fully deterministic, no randomness.

Conclusion

The pipeline delivers a complete, evidence-grounded bidding system. LightGBM was selected over all alternatives based on non-linear pricing relationships identified in EDA, its leaf-wise growth advantage over XGBoost, and native quantile objective enabling a three-model uncertainty architecture. Optuna TPE tuning delivered a 31% RMSE improvement. The agent combines auction theory (winner’s curse bid shading), quantitative finance (fractional Kelly with bankroll ratio), and information design (hyperbolic decay obscuring budget state from rivals). Every threshold — the 0.139 confidence boundary, the 0.414 pass limit, the 5% and 10% margin requirements — is grounded directly in EDA’s CV structural break findings.

Appendix: Key Formulas

$$(A1) \text{ Log-IQR bound} = \exp(Q_3(\ln p) + 1.5 \cdot \text{IQR}(\ln p)) - 1$$

$$(A2) \text{ Target encoding} = \hat{\mu}_c = (n_c \bar{y}_c + 10 \bar{y}_{\text{global}}) / (n_c + 10)$$

$$(A3) \text{ Uncertainty} = (\hat{p}_{0.75} - \hat{p}_{0.25}) / \hat{p}_{0.50}$$

$$(A4) \text{ Kelly fraction} = f_{\text{safe}} = \min(e/w^2 \cdot \frac{1}{2} \cdot B/B_0, f_{\text{max}})$$

$$(A5) \text{ Bid sequence} = b_{r+1} = b_r + \max((c_{\text{eff}} - b_r) \cdot f_{\text{safe}} \cdot (1 - 0.3w) / (1 + 0.5r), \$50)$$